Project SunSat: Project Enki

Critical Design Review

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Abstract

Second year computer system / robotics engineers, Rohaan Khawaj, Marcus Tsui & Leo Peintre, propose the development of a CubeSat payload designed for the acquisition of magnetic field data for regional crustal mapping. The authors aim to build upon prior experience as avionics engineers in Project SunSat. The project aims to design, prototype, and evaluate a payload capable of collecting low-noise, geomagnetic field data from stratospheric altitudes (30-35km) via a weather balloon. The collected data will be suitable for post processing and geological analysis. The scope of this project includes onboard data handling, telemetry to a ground control station, and data processing for use in geological studies. The primary outcome of this initial proposal is to communicate our project to TBD.

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